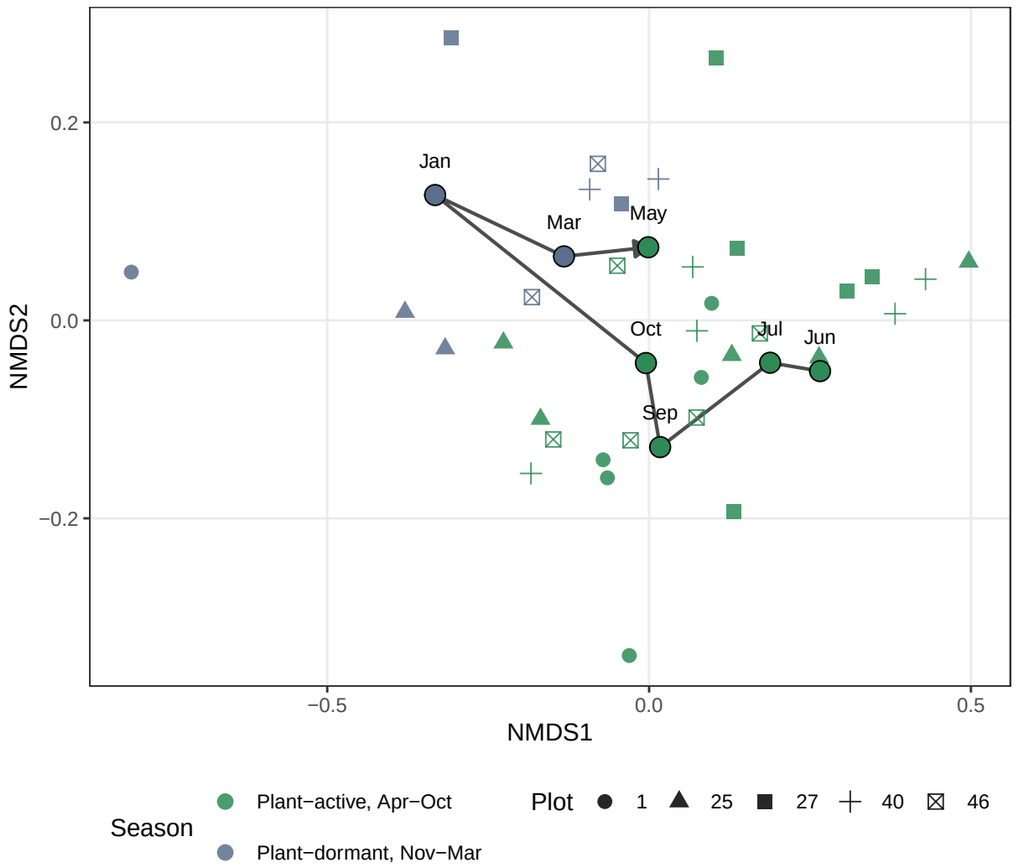
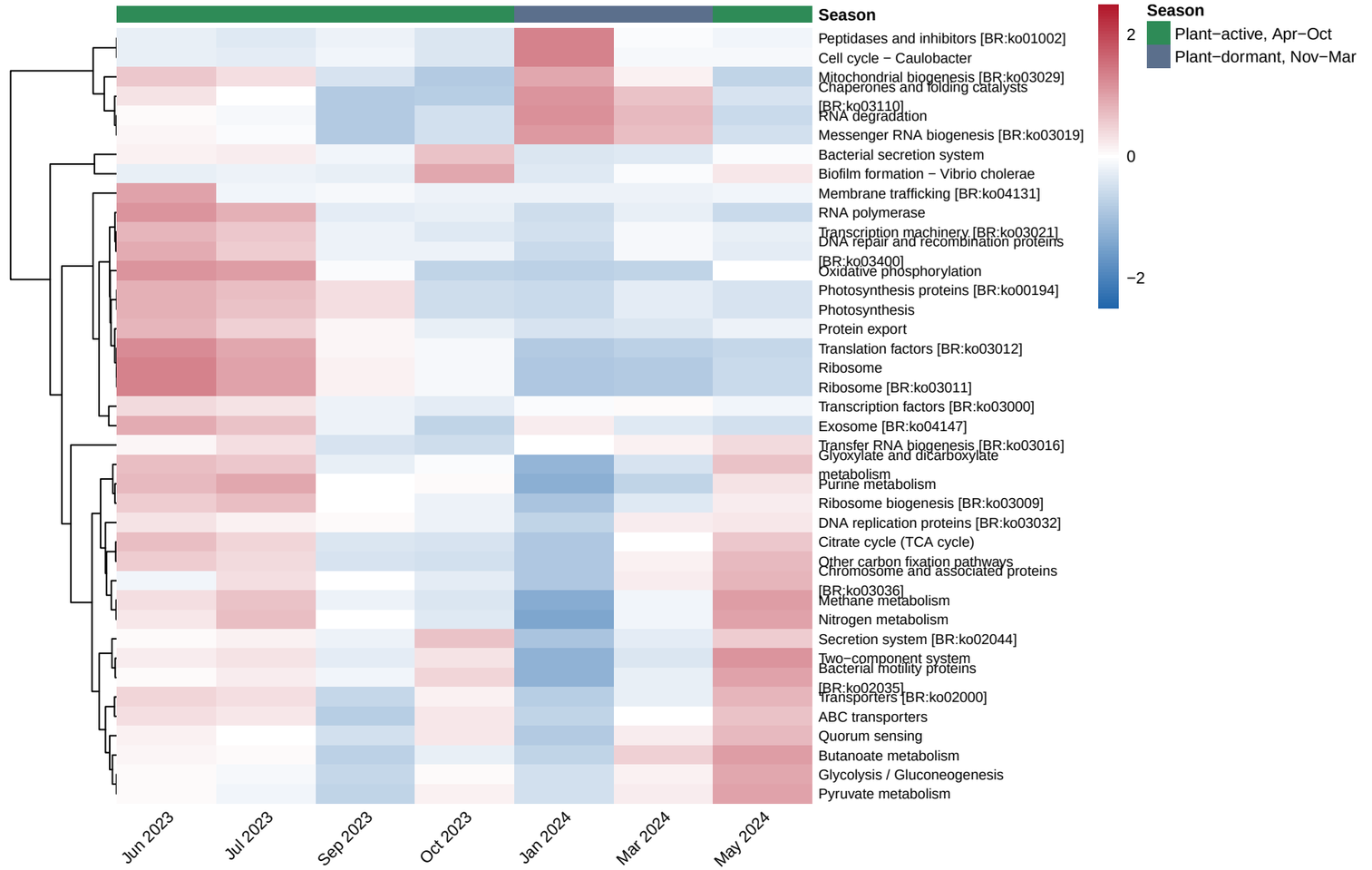
A

BRITE C Bray−Curtis NMDS; stress = 0.089
PERMANOVA R^2 = 0.234, F = 10.09, p = 0.001; dispersion p = 0.92

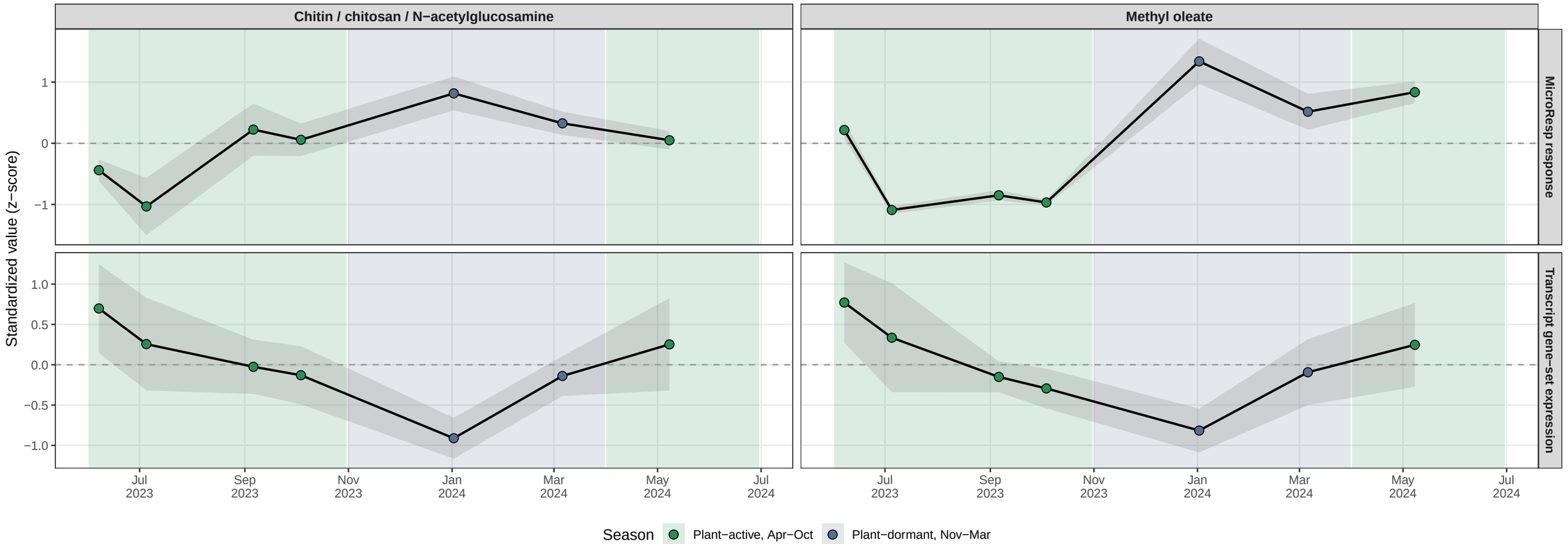


B Major BRITE C functions (expression z−score)



C Contrasting seasonal trajectories

Biologically curated before temporal analysis; aligned date means and +/− SE are shown on separate response rows



Panel B: the 40 functions were selected by abundance and prevalence, not temporal significance. Panel C: contrasting seasonal trajectories for pre−curated substrate families and transcript gene sets. Exploratory date−level correlations (n = 7 per family): chitin family Pearson r = −0.77 (p = 0.042); Spearman rho = −0.93 (p = 0.0025); methyl oleate Pearson r = −0.24 (p = 0.599); Spearman rho = −0.29 (p = 0.5345). These correlations are descriptive and do not establish mechanistic linkage.